UNIVERSITY OF THE ARMED FORCES ESPE

Object Oriented Programming

SOFTWARE ENGINEERING

NRC:30812

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Names Of The Members:

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Code Breakers

Artifacts reviewed:

- Product backlog
- Use case diagrams
- Class diagrams
- Code

Review type: Walkthrough Date: May 13, 2026

Findings In The Product Backlog

- Tasks too general: The product backlog contains very broad tasks that are not broken down into specific, actionable features.
  - Correction: Split each general task into smaller, more specific user stories (e.g., instead of "authentication", create "login", "logout", "password recovery").
- Lack of feature specificity: Features are not detailed enough for development.
  - Correction: Add acceptance criteria and technical notes to each feature.

Findings In Use Cases Diagram

- Missing actor: Only one actor (Teacher) is present, but more actors are needed (e.g., Student, Administrator).
  - Correction: Add at least Student and Administrator actors.
- Missing use cases:
- Account creation and login are not represented.
- Teacher must be able to view performance alerts.
- Teacher can also view educational materials.
- Incorrect use case: "Log participation timestamp" is not a valid use case (it is an action, not a functional requirement).
  - Correction: Replace with "Record participation" or remove it.
- Poor diagram layout: The distribution of elements makes the diagram hard to read.

Correction: Rearrange the layout, group related use cases, and use system boundaries clearly.

Findings In Class Diagram

- Only one diagram: The project shows only one class diagram, which is insufficient for a complete design.
  - Correction: Create separate diagrams for each major package (e.g., User Management, Content Management).
- Poor distribution and missing packages: Classes are not organized into packages.
  - Correction: Group classes into packages like model, view, controller, utils.
- Associations vs dependencies: The diagram uses dependencies where associations should be used.
  - Correction: Replace dependency arrows with association lines where classes have structural relationships.
- Missing multiplicity: Multiplicity (e.g., 1, 0.., 1..) is not defined.
  - Correction: Add multiplicity to all associations.
- Missing class: A "Lower" class is mentioned but not present.
  - Correction: Define the Lower class or remove it if unnecessary.

- Incorrect data types:
- "career" in Teacher class is defined as int, but should be String.
- Missing methods:
- viewGrades() and provideFeedback() are not included in the Teacher class.
    - Correction: Add these methods with appropriate parameters and return types.

Findings In The Code

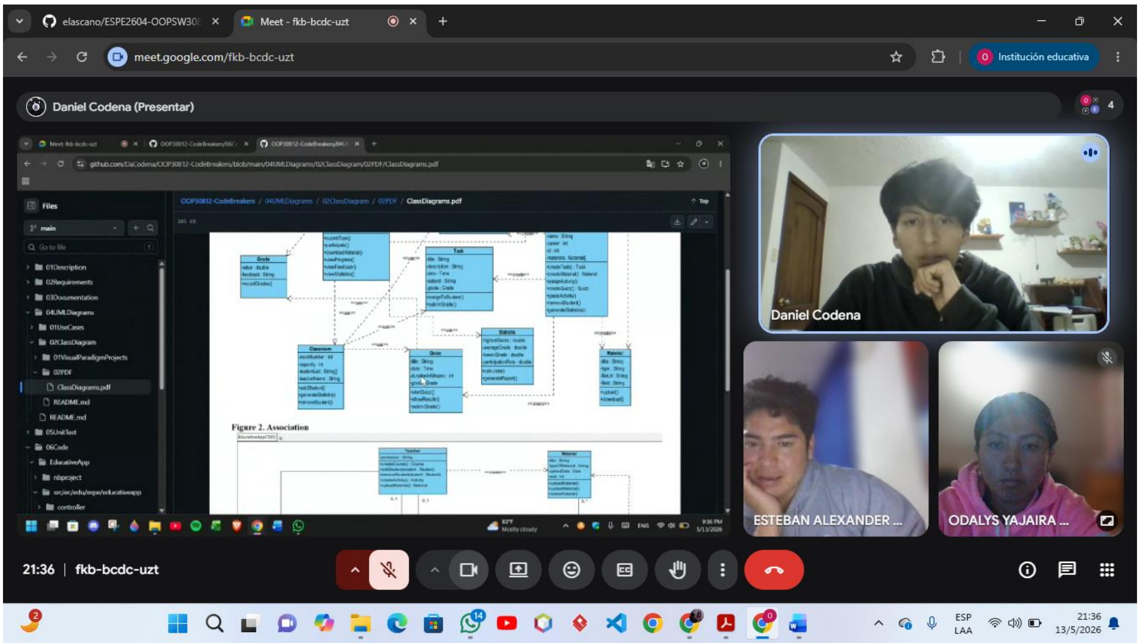
- Observed Issues:
- Class Material: Contains unnecessary/redundant comments (e.g., "return the title" for getTitle()). Correction: Remove obvious comments, keep only those explaining complex logic.
- Class Teacher: Method getTitle() has a comment that adds no value. Correction: Delete redundant comment.
- Class EducativeApp: Does not contribute much; only creates a menu that already exists elsewhere. Correction: Merge with main application class or remove duplication.

Additional Observations:

- Some classes have too many attributes, while others are missing attributes defined in the class diagram.
- Recommendation: Verify each class against the class diagram to ensure consistency.
- Comments are sometimes redundant or misleading.
- Recommendation: Use comments only to explain why, not what.

Summary of Recommendations

- Product Backlog: Break tasks into smaller, specific features.
- Use Case Diagram: Add missing actors (Student, Admin), missing use cases (login, view materials, view alerts), and fix layout.
- Class Diagram: Add packages, multiplicities, associations (not dependencies), missing methods, and correct data types.
- Code: Remove redundant comments, ensure class-diagram consistency, eliminate duplicated menu logic.


elascano/ESPE2604-OOPSW30
Meet - fkb-bcdc-uzt
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Daniel Codena (Presentar)
Figure 2. Association
Daniel Codena
ESTEBAN ALEXANDER ...
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